

ScaleNet: Learnable Multi-Scale Voxelization Architecture

1. Input Point Cloud

Point Cloud: $P \in \mathbb{R}^{N \times 4}$

N points: (x, y, z, r)

r = reflectance intensity

Feed

2. Learnable Voxel Scales (PhD Contribution)

Scales: $\theta = [\theta_1, \theta_2, \theta_3]$

Initialized: $[0.05m, 0.10m, 0.20m]$

Learned via gradient descent

`nn.Parameter (requires_grad=True)`

Use

Scale Assignment Network (ScaleNet)

MLP: $(x, y, z, r) \rightarrow [64 \rightarrow 32 \rightarrow 3]$

Gumbel-Softmax: $\sigma = \text{softmax}(z/\tau)$

Output: $\sigma_i \in \mathbb{R}^3$ (per point)

Differentiable scale selection

Gradient Flow

Backpropagation

$\partial L / \partial \theta$

$\partial L / \partial z_i$

$\partial L / \partial \sigma_i$

$\partial L / \partial f_i^{(s)}$

$\partial L / \partial f_i$

4. Multi-Scale Voxelization (3 Parallel Branches)

Fine Scale

θ_1 (~0.04m)

Voxel Grid 1

Resolution: Dynamic

Medium Scale

θ_2 (~0.12m)

Voxel Grid 2

Resolution: Dynamic

Coarse Scale

θ_3 (~0.25m)

Voxel Grid 3

Resolution: Dynamic

5. Scale-Specific Voxel Feature Encoding (VFE)

VFE Branch 1

PointNet-style: $f^{(s)}$

Output: Feature Map 1

VFE Branch 2

PointNet-style: $f^{(s)}$

Output: Feature Map 2

VFE Branch 3

PointNet-style: $f^{(s)}$

Output: Feature Map 3

6. Weighted Feature Aggregation

Weighted Sum: $f_i = \sum_{s=1}^3 \sigma_i^{(s)} \cdot f_i^{(s)}$

Differentiable w.r.t. both σ and θ

Adaptive per-point scale selection

End-to-end trainable

7. 3D Object Detection Head

Bounding Box Regression + Classification

Output: Boxes + Scores

Loss: Classification + Regression

KEY INNOVATIONS:

- ★ Learnable voxel scales (PhD Contribution)
- ★ Gumbel-Softmax for differentiable selection
- ★ Per-point adaptive scale assignment
- ★ End-to-end training with detection loss
- ★ No manual tuning required

PERFORMANCE:

- AP (Car, Moderate):
- Baseline: 70.21%
 - VoxAdapt: 85.09%
 - Improvement: +21.2%

Learned Scales:

- Fine: 0.041m
- Medium: 0.118m
- Coarse: 0.251m